

## Full Length Article

# Polyaniline-coated magnetic diatomite nanoparticles as a matrix for immobilizing enzymes



Mariana Paola Cabrera<sup>a,\*</sup>, Taciano França da Fonseca<sup>a</sup>, Raquel Varela Barreto de Souza<sup>a</sup>,  
Caio Rodrigo Dias de Assis<sup>a</sup>, Justiniano Quispe Marcatoma<sup>b,c</sup>, Jackeline da Costa Maciel<sup>d</sup>,  
David Fernando Morais Neri<sup>e</sup>, Fernando Soria<sup>f</sup>, Luiz Bezerra de Carvalho Jr.<sup>a,\*</sup>

<sup>a</sup> Departamento de Bioquímica e Laboratório de Imunopatologia Keizo Asami, Universidade Federal de Pernambuco, Cidade Universitária, 50670-901 Recife, PE, Brazil

<sup>b</sup> Facultad de Ciencias Físicas, Universidad Nacional Mayor de San Marcos, Ap-Postal 14-0149, Lima, Peru

<sup>c</sup> Centro de Investigaciones Tecnológicas, Biomédicas y Medioambientales (CITBM), Calle José Santos Chocano 199, Bellavista, Callao, Peru

<sup>d</sup> Universidade Federal de Roraima, Campus Paricarana, 69301-000 Boa vista, RR, Brazil

<sup>e</sup> Universidade Federal do Vale do São Francisco, Campus Petrolina, 56304-917 Petrolina, PE, Brazil

<sup>f</sup> Instituto de Investigaciones para la Industria Química, Universidad Nacional de Salta-CONICET, Buenos Aires N° 177, 4400 Salta, Argentina

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## ABSTRACT

This work proposes a simple, cost-effective, and efficient preparation of a composite made from magnetic diatomaceous earth (mDE) coated with polyaniline (mDE@PANI). The material was used as a matrix for immobilizing industrial enzymes: invertase, β-galactosidase, and trypsin. The mDE@PANI was characterized by several methods, and the results suggested that the nanoparticles were approximately 12 nm in size, exhibited superparamagnetic behavior, and displayed a good magnetic response and that magnetite comprised the main iron oxide phase. Moreover, several studies were conducted for all immobilized derivatives, including determination of optimal pH and temperature; kinetic parameters; thermal stability and reusability. The obtainment of three novel magnetic biocatalysts with superior performance (in terms of activity and stability) compared to their free counterparts demonstrated the efficacy of the mDE@PANI nanoparticles. In addition, the enzymatic derivatives can be easily recovered from the reactor by using an external magnetic field. Finally, the present methodology allowed the achievement of good mDE@PANI matrix together with three promising magnetic biocatalysts with several potential biotechnological applications.

## 1. Introduction

Enzymes are important biological macromolecules widely used in industrial applications. Their immobilizations confer advantages, such as improved thermal stability and reusability. The literature is replete with studies about the immobilization of enzymes on different matrices [1,2]. It is well known that the matrix has significant effects on the performance of the catalyst. Preparation of a composite material comprising two or more organic and/or inorganic compounds could be an interesting approach toward developing a matrix for biomolecule immobilization. Among inorganic materials, diatomaceous earth (DE), a lightweight mineral clay composed mainly of amorphous hydrated silica (SiO<sub>2</sub>·nH<sub>2</sub>O), has attracted much attention due to many of its properties, such as high cation exchange capacity, large surface area, porous structure, chemical inertness, low thermal conductivity, low cost, and ready availability [3]. DE (natural or modified) has been

explored in several biotechnological applications, with the removal of heavy metals [4–11] being the main application reported in the literature. However, more recent studies have described the use of this material as a potential functional drug carrier [12–15] and as a new poultry vaccine adjuvant [16]. Recently, Chen et al. [17] have reported the wide potential applications of functional nanocomposites produced from both magnetite and this type of material (mineral clay). In addition, some researchers have produced electrically conducting diatomite by coating it with polyaniline (PANI), one of the most-studied conducting polymers [18–21]. To the best of our knowledge, little attention has been paid to modified DE as a matrix for biomolecule immobilization [22–26]. PANI is a material highlighted by its direct electron transfer capability with immobilized enzymes [27]. The effects of PANI on catalytic performance of enzymes have been related to the immobilization methods. For adsorptive interactions, repercussions in enzymatic activity were reported as a consequence of the change on the

\* Corresponding authors at: Laboratório de Imunopatologia Keizo Asami (LIKA), Universidade Federal de Pernambuco, Cidade Universitária, Recife, PE CEP 50670-901, Brazil (L.B. de Carvalho).

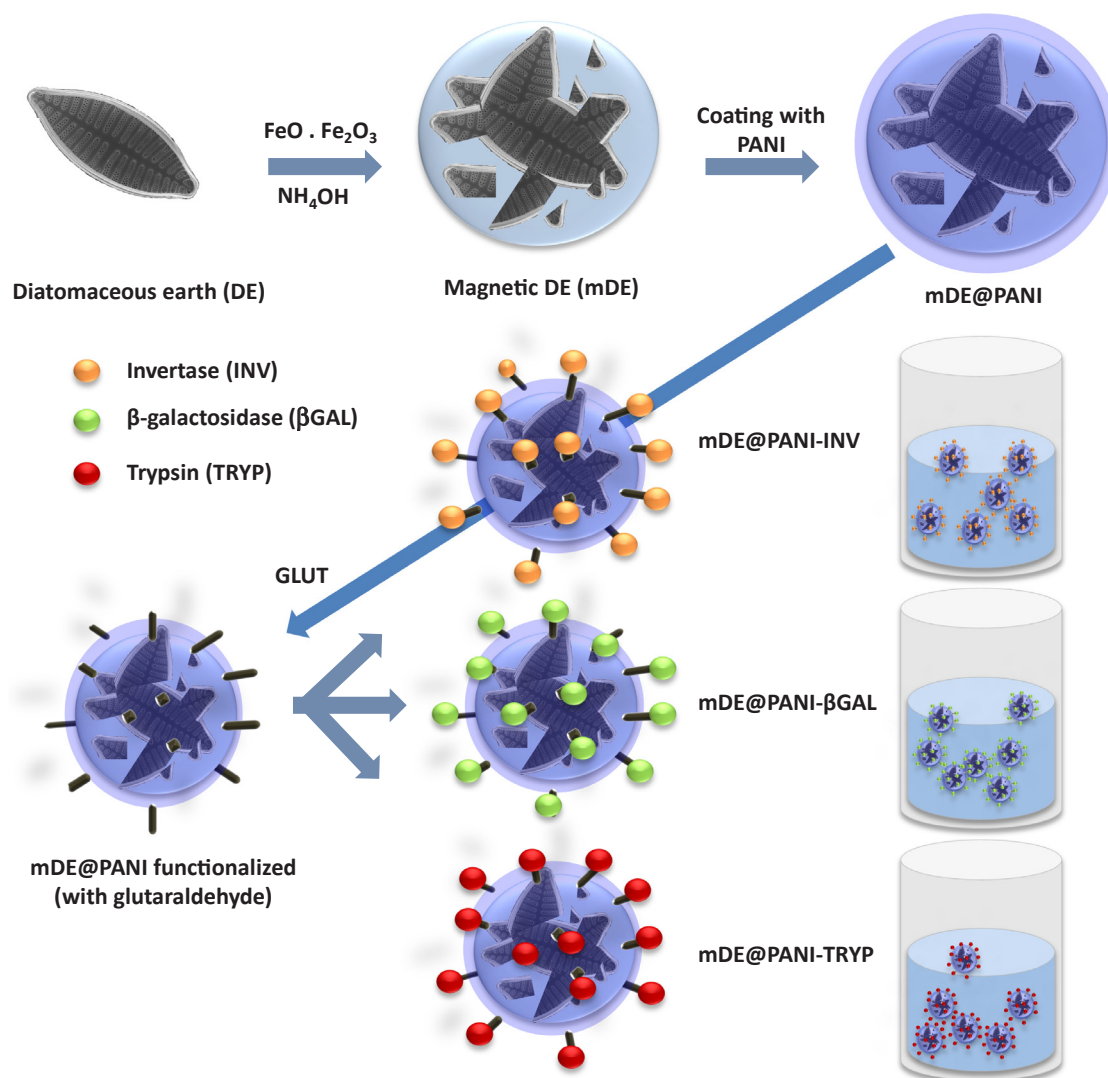
E-mail addresses: [marian.br09@gmail.com](mailto:marian.br09@gmail.com) (M.P. Cabrera), [lbcj.br@gmail.com](mailto:lbcj.br@gmail.com) (L.B.d. Carvalho).

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**Scheme 1.** Chemical strategy applied for the synthetic preparation of mDE@PANI nanoparticles and the three magnetic biocatalysts obtained from them.

enzyme active site after immobilization process [28]. And for covalent immobilization, intermediate oxidation states of PANI implied a greater amount of glutaraldehyde bound which, in turn, allowed better activity retention when compared with the fully oxidized or reduced states of the conducting material [29]. Here, PANI coating on magnetic DE (mDE) added chemical groups ( $-\text{NH}_2$ ) on magnetic surface allowing covalent binding of enzymes. Therefore, the mDE coated with PANI (mDE@PANI) followed the chemical strategy (Scheme 1) to covalently immobilize three industrial enzymes: invertase (EC 3.2.1.26),  $\beta$ -galactosidase (EC 3.2.1.23), and trypsin (EC 3.4.21.4). Invertase is mostly used to convert sucrose to invert sugar (glucose and fructose) that is an important product in the food and beverage industries. In our laboratory invertase has been previously immobilized on magnetic DE (mDE) [30,31]. However, the enzyme was covalently immobilized on mDE functionalized by a silane agent (APTES – aminopropyltriethoxysilane). Another enzyme of particular interest is  $\beta$ -galactosidase that hydrolyzes lactose to glucose and galactose in dairy products. This is important  $\beta$ -galactosidase application for the production of free lactose foods aiming people suffering from lactose intolerance. Furthermore, production of galacto-oligosaccharides (non-digestible carbohydrate) from lactose by  $\beta$ -galactosidase immobilized on magnetic materials has been widely explored [32–38]. Trypsin also has many different industrial and biomedical applications including, in particular, as a laboratory reagent for proteomic analysis [39–42].

The major contribution of this work is the employment of the mDE@PANI nanoparticles as an efficient and attractive matrix for covalent immobilization of the enzyme. Several physical–chemical techniques were used to characterize the mDE@PANI as well as some properties of the three novel biocatalysts were studied: optimal pH and temperature; kinetic parameters; thermal stability and reusability.

## 2. Materials and methods

### 2.1. Materials

DE was kindly donated by TAMER Company (Salta, Argentina). Invertase (Baker's yeast),  $\beta$ -galactosidase (*Aspergillus oryzae*), trypsin (porcine pancreas), potassium permanganate, aniline, glutaraldehyde, bovine serum albumin, lactose, N- $\alpha$ -benzoyl-DL-arginine-p-nitroanilide (BAPNA), and dimethylsulfoxide (DMSO) were purchased from Sigma Aldrich Chemicals (St. Louis, USA). All other chemicals were of the highest purity available commercially.

### 2.2. Preparation of mDE@PANI nanoparticles (mDE@PANI<sub>nano</sub>)

The mDE was prepared according to the procedure described by Cabrera et al. [31]. Before the PANI-coating process, mDE was washed

with sufficient distilled water until the pH became neutral and was recovered using a magnetic field (Ciba Corning; 0.6 T). Coating was achieved by oxidative polymerization of aniline on the mDE surface using potassium permanganate as an oxidizing agent. Briefly, mDE nanoparticles (0.25 g) were treated with 0.1 M  $\text{KMnO}_4$  (1 mL) in aqueous solution at 25 °C for 1 h. Subsequently, 0.25 M aniline (1 mL) solution prepared in 2.0 M HCl was added to the magnetic suspension. The polymerization process was performed in the dark at 4 °C for 30 min. It is important to note that after each step of the polymerization process, magnetic nanoparticles were successively washed (fifteen times) with distilled water to remove any residual reagent. Finally, the mDE@PANI<sub>nano</sub> were dried at 50 °C for 4 h.

### 2.3. Physical–chemical characterization of mDE@PANI<sub>nano</sub>

XRD was performed using a Siemens D5000 X-ray diffractometer with  $\text{CuK}\alpha$  radiation ( $\lambda = 1.5406 \text{ \AA}$ ). SEM-EDS was performed using a scanning electron microscope (FEI Model QUANTA 200 FEG). Specific surface area and porosity were determined using a Micromeritics ASAP 2420 porosimeter. Isotherms were determined at 77 K using  $\text{N}_2$  as an adsorbate. The Brunauer–Emmett–Teller model and the Barrett–Joyner–Halenda-plot method were employed to determine the specific surface area and porosity parameters (pore size distribution and pore volume). FTIR spectra were recorded ( $4000\text{--}400 \text{ cm}^{-1}$  range) in a model IFS 66 spectrometer (Bruker). Mössbauer spectra were obtained at 4.2 K in a transmission geometry using a conventional  $^{57}\text{Fe}$  Mössbauer spectrometer employing a 50 mCi  $^{57}\text{Co}/\text{Rh}$  source. The spectra were analyzed using the least squares method assuming Lorentzian line shapes. The isomer shift ( $\delta$ ) values are relative to  $\alpha\text{-Fe}$  at room temperature [30]. To evaluate the magnetic property of samples, various magnetic fields from 0 to 50 kOe (5.0 T) were applied at 298 K using a SQUID magnetometer (Quantum Design Model MPMS-5S).

### 2.4. Enzyme immobilization

All immobilized derivatives were obtained by covalent attachment using glutaraldehyde, as a chemical arm, for immobilization of the enzyme (invertase,  $\beta$ -galactosidase and trypsin) onto the mDE@PANI<sub>nano</sub>. Briefly, the magnetic support (0.01 g) was activated with 2.5% v/v glutaraldehyde (2 mL) for 1 h under stirring at 25 °C. The activated mDE@PANI<sub>nano</sub> were washed several times with distilled water until the washings became colorless. The enzyme solution (1 mL) was incubated with activated mDE@PANI<sub>nano</sub> (0.01 g) for 12 h at 4 °C with mild stirring. Each enzyme solution was prepared as follows: invertase (0.1 mg protein  $\text{mL}^{-1}$ ) prepared in 0.2 M sodium acetate buffer (pH 5.0);  $\beta$ -galactosidase (0.2 mg protein  $\text{mL}^{-1}$ ) prepared in 0.1 M citrate–phosphate buffer (pH 4.5), and trypsin (0.1 mg protein  $\text{mL}^{-1}$ ) prepared in 0.1 M Tris–HCl buffer (pH 8.0).

After immobilization process, the mDE@PANI-enzyme preparations were washed several times with the appropriate enzyme buffer. The supernatant and washings were used for protein determination according to Lowry et al. [43] using bovine serum albumin as a standard. The mDE@PANI-immobilized invertase (mDE@PANI-INV), mDE@PANI-immobilized  $\beta$ -galactosidase (mDE@PANI- $\beta$  GAL), and mDE@PANI-immobilized trypsin (mDE@PANI-TRYP) were stored in the appropriate enzyme buffer at 4 °C until further use.

### 2.5. Enzyme assay

Invertase activity was determined using 0.15 M sucrose (10 mL) prepared in sodium acetate buffer (0.2 M, pH 4.5) as a substrate. After 15 min of incubation at 25 °C, 20  $\mu\text{L}$  of the sample was withdrawn and added to 2.0 mL of working solution to measure released glucose using a glucose oxidase–peroxidase enzymatic kit (Dole, Goiás, Brazil).

$\beta$ -galactosidase activity was measured using lactose (1 mL) prepared in citrate–phosphate buffer (0.1 M, pH 5.0) as a substrate. The

released glucose was measured and determined by the same way that the invertase activity. Therefore, the enzymatic activity unit (U) for both invertase and  $\beta$ -galactosidase was defined as the amount of enzyme able to release 1  $\mu\text{mol}$  of glucose per minute under these assay conditions.

Trypsin activity was determined according to Amaral et al. [44] using BApNA prepared in DMSO as a chromogenic substrate. One unit (U) of trypsin was defined as the amount of enzyme that can hydrolyze 1  $\mu\text{mol}$  of BApNA per minute under these assay conditions.

### 2.6. Biochemical characterization

#### 2.6.1. pH and temperature profiles

Invertase activity was assayed over a pH range of 3.0–7.0 (0.2 M sodium acetate buffer for pH 3.0–5.5 and 0.2 M sodium phosphate buffer for pH 5.5–7.0) at 25 °C for 15 min. For the temperature profile study, the activity was assayed at different temperatures (30–70 °C) for 15 min at pH 4.5.

For  $\beta$ -galactosidase, the effect of pH on the enzyme activity was investigated in the range 3.0–7.0 (0.1 M citrate–phosphate buffer) at 25 °C for 10 min. The temperature profile was examined between 30 °C and 80 °C.

To determine the optimum pH of trypsin, 0.1 M Tris–HCl buffer was used for pH 6.5–9.5 and 0.2 M Glycine–NaOH for pH 10–12 at 25 °C for 15 min. The temperature profile was examined between 30 °C and 70 °C.

#### 2.6.2. Kinetic parameters

The hydrolytic activities of free and immobilized enzymes were assayed at 25 °C using different final concentrations of the substrate depending on the enzyme: 25–250 mM sucrose (invertase), 29–1462 mM lactose ( $\beta$ -galactosidase), and 0.50–20 mM BApNA (trypsin). The kinetic parameters, including  $k_m$  and  $V_{max}$ , were obtained by non-linear regressions using the PRISM 5.0® software (GraphPad, USA). Statistical analysis of the parameters was performed by means comparison test (student's T-test) after meeting normality and equal variances requirements using Origin 8.0® software (Microcal, USA).

Assays were performed at increasing temperatures up to the optimum temperature, i.e., from 30 °C to 50 °C (invertase and trypsin) and from 30 °C to 60 °C ( $\beta$ -galactosidase), to determine the activation energy (AE) using the Arrhenius plot [ $\ln(k)$  vs  $1/T$ ] and the logarithmic variant of the Arrhenius equation:

$$\ln(k) = \ln(A) - \frac{AE}{R} \times \frac{1}{T} \quad (1)$$

where  $k$  = rate constant,  $A$  = pre-exponential factor,  $AE$  = activation energy,  $R$  = gas constant, and  $T$  = absolute temperature.

Another kinetic parameter evaluated was the rate enhancement caused by the enzyme in relation to the spontaneous reaction [45] using the following equation:

$$k = \left( \frac{kT}{h} \right) \times e^{\left( -\frac{AE}{R} \right)} \quad (2)$$

where  $k$  = rate constant,  $k$  = Boltzmann constant,  $T$  = absolute temperature,  $h$  = Planck's constant,  $AE$  = activation energy, and  $R$  = gas constant.

#### 2.6.3. Thermal stability

Enzymes were incubated at different temperatures under stirring for varying amount of time (30–120 min). The temperature ranges used were 45–55 °C (invertase prepared in 0.2 M sodium acetate buffer, pH 5.0), 55–65 °C ( $\beta$ -galactosidase prepared in 0.1 M citrate–phosphate buffer, pH 4.5), and 45–60 °C (trypsin prepared in 0.1 M Tris–HCl buffer, pH 8.0). Treated samples were cooled to 25 °C, and then their enzymatic activity determined as described in Section 2.5 considering the first one as 100%.

## 2.7. Reusability

The three immobilized derivatives were stored in the appropriate enzyme buffer at 4 °C and were reused 10 times at 30-min interval. After each assay, the enzymatic preparation was washed fifteen times with the appropriate enzyme buffer, and a new substrate solution was added for the next activity cycle. The residual activity (%) was determined by measurement of enzyme activity as reported in Section 2.5 considering the first one as 100%.

## 3. Results and discussion

Here three novel magnetic biocatalysts were obtained by covalent binding between the mDE@PANI<sub>nano</sub> and the enzyme (invertase,  $\beta$ -galactosidase, and trypsin). Due to inorganic nature of the mDE nanoparticles, firstly we carried out the PANI coating which provided amine groups ( $-\text{NH}_2$ ) and a hydrophilic environment on the magnetic surface (mDE). Then, such covalent binding between the enzyme ( $-\text{NH}_2$ ) and the mDE@PANI<sub>nano</sub> ( $-\text{NH}_2$ ) was successfully performed by addition of glutaraldehyde used as a chemical arm for enzyme immobilization. The Scheme shows the chemical strategy used for preparation of mDE@PANI<sub>nano</sub> and the immobilized derivatives. The immobilization yield (%) for the magnetic biocatalysts: mDE@PANI-INV, mDE@PANI- $\beta$ GAL and mDE@PANI-TRYP were around of 20, 10 and 47%, respectively. It is important to mention that these percentage values above are related to the following enzyme concentrations: 0.1 mg protein mL<sup>-1</sup> (invertase), 0.2 mg protein mL<sup>-1</sup> ( $\beta$ -galactosidase) and 0.1 mg protein mL<sup>-1</sup> (trypsin), which constituted the best combination of specific and enzymatic activities. Physical-chemical characterization of the magnetic nanoparticles together with the main biochemical features of the immobilized derivatives are presented below.

### 3.1. Physical-chemical characterization of mDE@PANI<sub>nano</sub>

#### 3.1.1. XRD

XRD analyses were used to further evaluate the phase identification and crystalline structures of the samples. XRD patterns of magnetite, mDE, and mDE@PANI are presented in Fig. 1. The analysis of the XRD pattern of magnetite allowed the measurement of the cell parameter,  $a = 0.835$  nm ( $\text{Fe}_3\text{O}_4$ ; ICDD N°. 19-0629) [46]. The XRD patterns of mDE and mDE@PANI (Fig. 1B-C) are in good agreement with that of a mixture of amorphous silica [4] and are characterized by a broad peak centered at  $2\theta = 21.8^\circ$  and the presence of a crystalline silicon oxide ( $\text{SiO}_2$ ; ICDD N°. 82-1409) and Albite ( $\text{NaAlSi}_3\text{O}_8$ ; ICDD N°. 09-0466). Analysis of the XRD patterns revealed that magnetite was the most predominant crystalline phase in all samples. Furthermore, the intensity of the magnetite peaks did not change after coating the magnetite/diatomite composite with PANI to form mDE@PANI<sub>nano</sub>. PANI also exhibited some degree of crystallinity and produced three peak maxima at approximately  $2\theta = 20^\circ$ ,  $25^\circ$ , and  $27^\circ$ , as previously reported by Feng et al. [47]. The XRD patterns of mDE and mDE@PANI were similar, indicating that PANI has no effect on the crystallization performance of the diatomite composite [19]. Jaramillo-Tabares et al. [48] have reported that oxidative polymerization is one of the most efficient techniques to effect good polymer-particle interaction. The coating process with PANI by oxidative polymerization of aniline did not markedly change the size of mDE@PANI<sub>nano</sub>, according to the XRD analysis. The grain size for each sample, as calculated from the broadening of the main XRD peak using the Scherrer formula, was found approximately 12 nm for mDE [30], representing a small increase of approximately 4% in grain size for mDE@PANI<sub>nano</sub>.

#### 3.1.2. Mössbauer spectroscopy

Mössbauer spectra of magnetic samples acquired at 4.2 K are shown in Fig. 2. The spectra were fitted using a two sextet model (sextet 1 and 2) for all samples [30]. For magnetite, the first sextet (sextet 1) had a

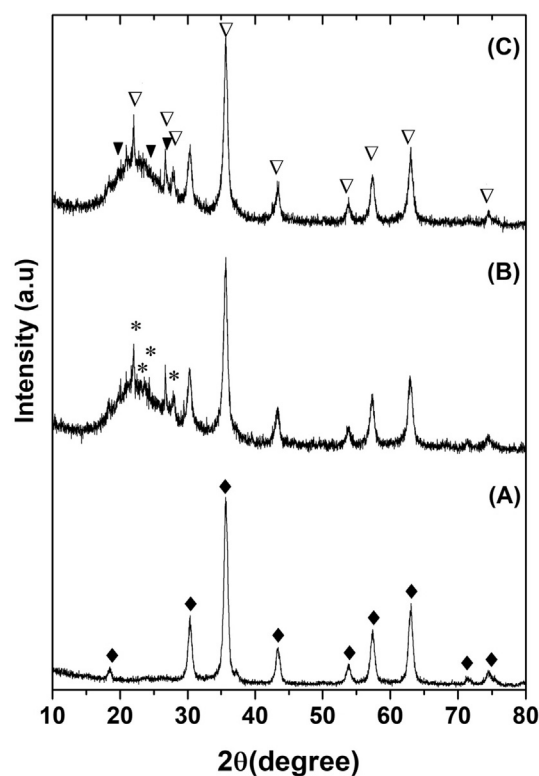


Fig. 1. XRD plots of (A) magnetite, (B) mDE and (C) mDE@PANI. (▽) Quartz, (\*) Albite, (♦) Magnetite.

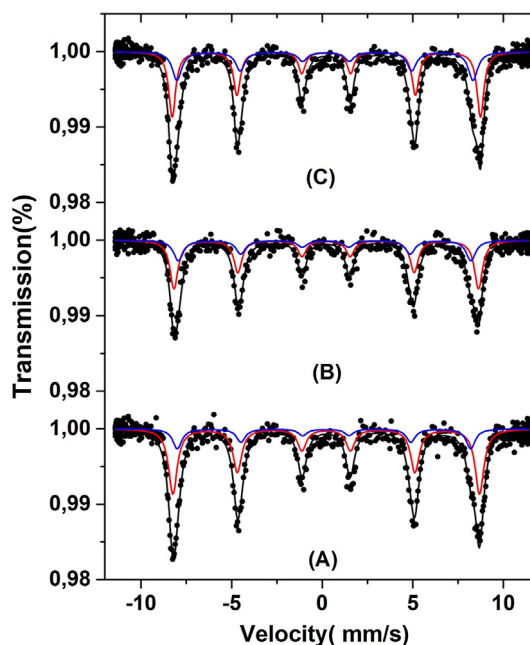


Fig. 2. Mössbauer spectra of (A) magnetite, (B) mDE and (C) mDE@PANI. Scattered points are data points, and the fitted spectrum is shown as the black line. The subspectra shown as red and blue lines are the component subspectra corresponding to sextet 1 and sextet 2, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

hyperfine magnetic field,  $B_{\text{hf}} = 52.5$  T, and an isomer shift,  $\delta = 0.33$  mm s<sup>-1</sup>, assigned to  $\text{Fe}^{3+}$  ions, whereas the second sextet (sextet 2) presented  $B_{\text{hf}} = 50.2$  T and  $\delta = 0.26$  mm s<sup>-1</sup> corresponding to mixed  $\text{Fe}^{2+}$ - $\text{Fe}^{3+}$  ions [49]. These values are similar to those of the bulk



**Table 1**

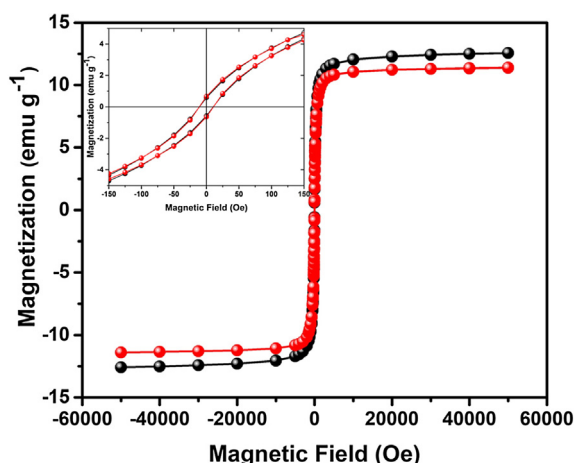
Mössbauer hyperfine parameters for studied samples.  $\delta$ : isomer shift;  $B_{\text{hf}}$ : hyperfine magnetic field; Area: relative area. Uncertainty in  $\delta$  is 0.02 mm/s, whereas that is  $B_{\text{hf}}$  is 0.5 T. Area is accurate within 2%

| Sample    | Component | $\delta$<br>(mm s <sup>-1</sup> ) | $B_{\text{hf}}$<br>(T) | Area<br>(%) |
|-----------|-----------|-----------------------------------|------------------------|-------------|
| Magnetite | Sextet 1  | 0.33                              | 52.5                   | 75.0        |
|           | Sextet 2  | 0.26                              | 50.2                   | 25.0        |
| mDE       | Sextet 1  | 0.33                              | 52.1                   | 63.0        |
|           | Sextet 2  | 0.27                              | 50.1                   | 37.0        |
| mDE@PANI  | Sextet 1  | 0.33                              | 52.7                   | 55.8        |
|           | Sextet 2  | 0.29                              | 50.7                   | 44.2        |

material [50]. A deviation in the ideal area ratio (1:2) of the iron in tetrahedral and octahedral positions obtained from the subspectra area could be associated with the presence of maghemite in the sample or to the smaller particle size compared with their bulk counterpart [51,52]. The hyperfine magnetic fields for mDE (sextet 1 = 52.1 T and sextet 2 = 50.1 T) and mDE@PANI (sextet 1 = 52.7 T and sextet 2 = 50.7 T) showed values similar to those of magnetite (Fig. 2B and C and Table 1). In contrast to observations reported by Jaramillo-Tabares et al. [48], the mDE and mDE@PANI spectra showed an increase in sextet 2 compared with the magnetite spectrum. This finding could represent a solid-state electronic interaction, such as an electron transfer process between the magnetic (magnetite present in the mDE nanoparticles) and non-magnetic (PANI) components. In addition, the results obtained showed that the PANI-coating process did not affect the iron oxides phases present in the mDE because the Mössbauer hyperfine parameters for the mDE@PANI<sub>nano</sub> showed no relevant changes. Mössbauer results allowed the conclusion that hematite ( $\alpha$ -Fe<sub>2</sub>O<sub>3</sub>) is not present in the magnetic samples and also suggested the presence of non-stoichiometric magnetite or a mixture of magnetite (Fe<sub>3</sub>O<sub>4</sub>) and maghemite ( $\gamma$ -Fe<sub>2</sub>O<sub>3</sub>), in which magnetite is the principal iron oxide phase.

### 3.1.3. Magnetization measurement

The magnetic properties of samples were measured by applying an external magnetic field at 298 K. The superparamagnetic behavior of mDE and mDE@PANI is illustrated in Fig. 3 (inset graph). It is important to note that the inset does not constitute a significant hysteresis loop due to the small values (close to zero) of remnant magnetization ( $M_r$ ) and coercivity ( $H_c$ ). In addition, saturation magnetization values ( $M_s$ ) were 12.4 and 11.3 emu g<sup>-1</sup> for mDE and mDE@PANI, respectively (Fig. 3). These results suggested that after PANI coating, no



**Fig. 3.** Magnetic properties of mDE (black balls) and mDE@PANI (red balls) at 298 K. The inset graph details the superparamagnetic behavior of the magnetic samples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

substantial change occurred in the magnetic properties of mDE. When these magnetization values were compared with those of magnetite (synthesized by the same methodology), a decrease of 2.5-fold was observed. Surface effects, such as a magnetically inactive layer due to the presence of non-magnetic matter (e.g., PANI), could be responsible for this decline [53], and similar results have also been reported for other magnetic composites [34,54]. Furthermore, the mDE@PANI<sub>nano</sub> exhibited a strong magnetic response after application of an external magnetic field (Ciba Corning; 0.6 T), and only a few seconds were required to separate the mDE@PANI<sub>nano</sub> from the reaction medium.

### 3.1.4. SEM-EDS and surface area parameters

SEM image of partial DE frustules before the magnetization process is shown in Fig. S1A (see Supplementary material). When this material was magnetized and then coated with PANI, physical changes to the DE were observed (Fig. S1B). The addition of chemical reagents (ammonium hydroxide, potassium permanganate, and aniline) coupled with temperatures of up to 100°C during the mDE@PANI<sub>nano</sub> synthesis process could have contributed to the destruction of DE frustules [21]. Thus, mDE@PANI<sub>nano</sub> were morphologically different with an appearance of irregular layers and rougher surfaces. Moreover, Fig. S1B also shows the pores of mDE@PANI<sub>nano</sub> partially filled with PANI after the coating process. A similar observation for non-magnetic DE has been reported by Li et al. [18] who observed that PANI polymerization process occurs at the pore surface of DE particles.

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.apsusc.2018.06.238>.

Surface parameter data can also help to corroborate some of the findings mentioned above with regard to the polymerization process. Cabrera et al. [31] have reported a 1.7-fold increase in specific surface area after magnetization of DE nanoparticles. In this study, we observed a slight decrease in specific surface area (< 10%) after PANI polymerization (Fig. S1C). Furthermore, the pore volume and mean pore size values also decreased compared with mDE. However, with a pore size of 8.5 nm, mDE@PANI<sub>nano</sub> exhibit mesoporous solid behavior (IUPAC-classification) similarly to mDE nanoparticles [31]. The EDS spectrum of mDE@PANI<sub>nano</sub> (Fig. S1D) was similar to that of mDE, with the only difference being the carbon peak caused by the presence of PANI.

It is worthwhile to highlight that any other non-magnetic components such as neat PANI spheres were removed from the final product (mDE@PANI<sub>nano</sub>) during the fifteen washes performed after the coating procedure. The mDE@PANI<sub>nano</sub> were only attracted by the external magnetic field (Ciba Corning; 0.6 T) in the washing process.

### 3.1.5. FTIR

The main absorption peaks of FTIR spectra of the magnetic samples are shown in Fig. 4. Two absorption bands characteristic of the Fe–O bond (magnetite) are exhibited at approximately 631 and 566 cm<sup>-1</sup> [55,56]. As reported by Jordan et al. [57], the weak band near 1638 cm<sup>-1</sup> for bare magnetite nanoparticles is probably ascribed to the N–H stretching mode (amine group). The three more common absorption bands of DE at 1097 cm<sup>-1</sup> (asymmetric stretching), 797 cm<sup>-1</sup> (symmetric stretching), and 470 cm<sup>-1</sup> (bending) can be observed in Fig. 4B [12,58]. Moreover, bands similar to the Fe–O bond were observed in the mDE and mDE@PANI<sub>nano</sub>, supporting the presence of the magnetic component (magnetite) in the nanoparticles. Because PANI and DE presented similar absorption bands, it was not possible to assign the corresponding wavenumber for each component in the mDE@PANI<sub>nano</sub> spectrum (Fig. 4C) [20]. Therefore, we believe that Fig. 4C depicts contributions from both components. The weak absorption band observed at 1482 cm<sup>-1</sup> is attributed to the C=C stretching mode for the benzenoid rings [20,59]. In addition, the SEM image (Fig. S1B, see Supplementary material) of the mDE@PANI<sub>nano</sub> corroborated that PANI formation had occurred on the magnetic DE surface (rugged surface), whereas the specific surface area results (Fig. S1C) suggested that the

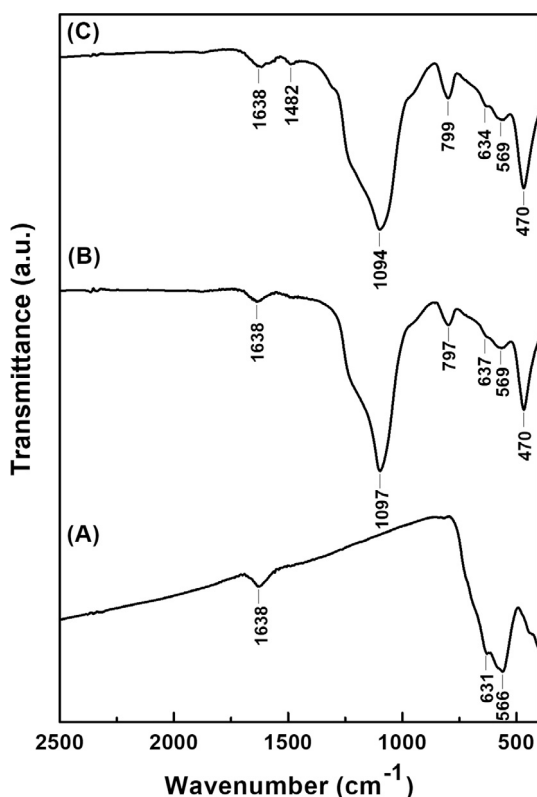


Fig. 4. FTIR spectra of the magnetic samples. (A) magnetite, (B) mDE, and (C) mDE@PANI.

mDE pores had been filled with PANI particles (decrease in pore size). Similar conclusions for modified PANI-diatomite particles were reported by Li et al. [18].

### 3.2. Biochemical characterization

#### 3.2.1. pH and temperature profiles

Several studies on invertase characterization have reported an optimal pH and temperature in the ranges of 4.0–5.5 and 45–65 °C, respectively [60–67]. Our results are presented in Table 2, and the properties of the immobilized derivative were not different from those of the free enzyme. These findings indicated that the immobilization process on mDE@PANI<sub>nano</sub> did not produce significant alterations regarding optimal pH and temperature under the experimental conditions. Furthermore, the lack of variation in optimal parameters could be explained by the strong covalent bonds that might also preserve the flexibility of the enzyme under pH and temperature changes. Some authors also described no changes in pH and temperatures profiles after invertase immobilization [23,63,67,68]. On the other hand, alterations in some of these parameters were detected for  $\beta$ -galactosidase and

Table 2

Properties and kinetic parameters of free enzymes and enzymes immobilized on mDE@PANI nanoparticles.

| Sample                 | pH  | T (°C) | $k_m$ (mM) | $V_{max}$ (U mg <sup>-1</sup> protein) | Rate enhancement <sup>a</sup> |
|------------------------|-----|--------|------------|----------------------------------------|-------------------------------|
| Invertase              | 4.5 | 50     | 73         | 630                                    | $2.24 \times 10^{11}$         |
| mDE@PANI-INV           | 4.5 | 50     | 199        | 3830                                   | $3.15 \times 10^{12}$         |
| $\beta$ -galactosidase | 4.5 | 60     | 80         | 443                                    | $7.69 \times 10^{16}$         |
| mDE@PANI- $\beta$ GAL  | 5.0 | 60     | 143        | 3903                                   | $7.24 \times 10^{14}$         |
| Trypsin                | 9.0 | 50     | 2          | 84                                     | $8.57 \times 10^{13}$         |
| mDE@PANI-TRYP          | 9.5 | 55     | 0.7        | 144                                    | $3.69 \times 10^{14}$         |

<sup>a</sup> According to Assis et al. [45].

trypsin immobilization on mDE@PANI<sub>nano</sub>. For free and immobilized  $\beta$ -galactosidase, we found the same optimum temperature but a slight increase in the pH (Table 2). Similar results have been reported for immobilized  $\beta$ -galactosidase from the same source [69,70]. Neri et al. [71] reported the same pH and temperature profile for the free and immobilized forms of *Kluyveromyces lactis*  $\beta$ -galactosidase. These authors also demonstrated that these parameters may vary depending on the enzyme source as well as the type of support and immobilization method.

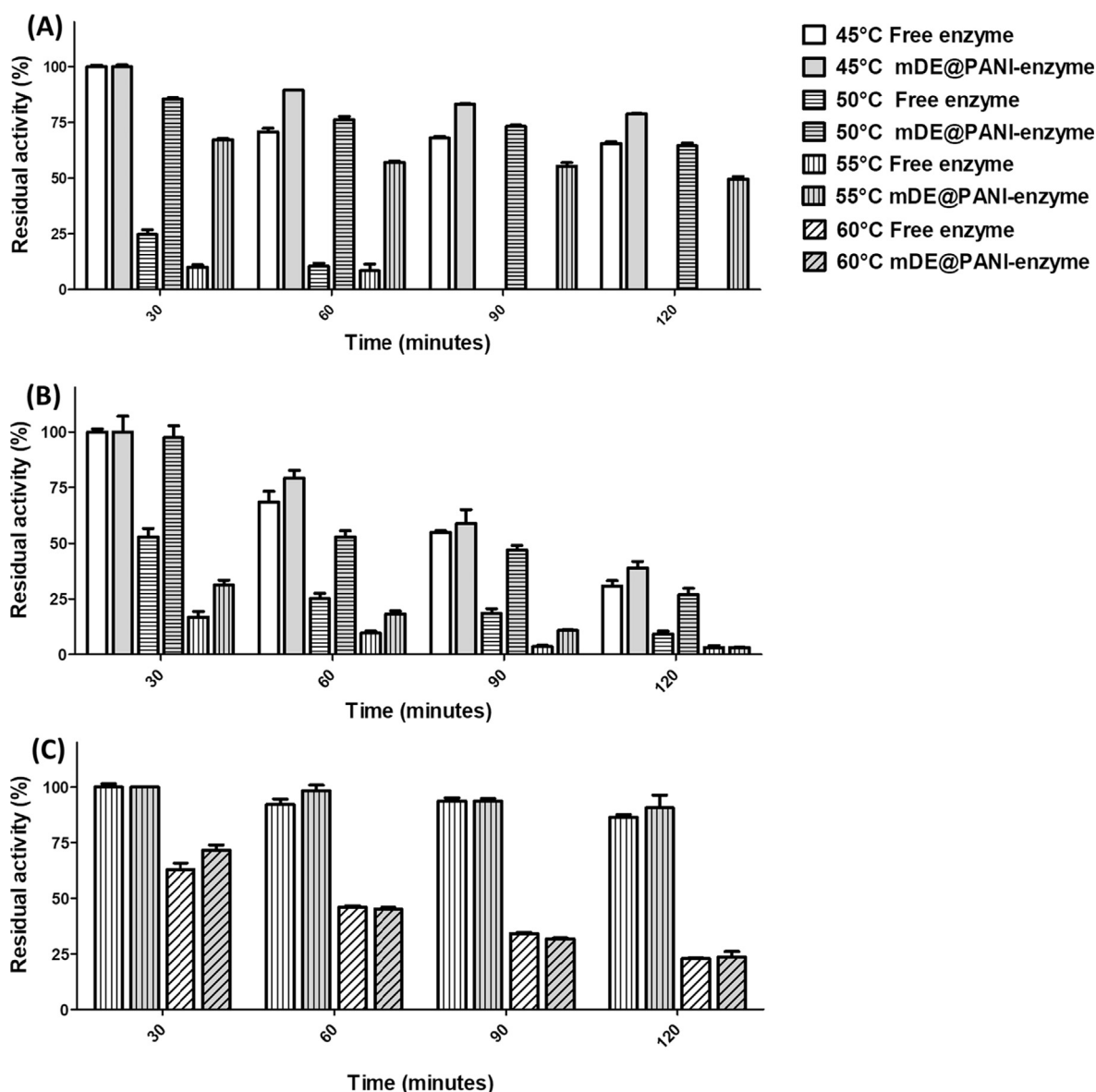
Table 2 also shows the effects of pH and temperature on BAPNA hydrolysis catalyzed by free and immobilized trypsin. After immobilization, increases of 0.5 and 5 °C in pH and temperature, respectively, were observed. Monteiro et al. [72] have reported similar results for free and immobilized trypsin on polysaccharide film. The immobilization process confers greater stability to the biomolecule (increased stiffness of protein structure) due to interactions between the enzyme and support surface. Here, we showed a modest increase in optimum temperature after immobilization, which is consistent with the findings of Bayramoglu et al. [73], who also reported an increase of 5 °C after trypsin immobilization onto magnetic beads.

#### 3.2.2. Kinetic parameters

The apparent kinetic parameters of free and immobilized enzymes are presented in Table 2. With the exception of trypsin, an increase in apparent  $k_m$  values after immobilization process was observed. The student's T-test ( $p < 0.05$ ) showed a statistically significant difference between apparent  $k_m$  of the two enzymatic preparations (free and immobilized enzyme). As reported by Altinok et al. [62] changes in the kinetic parameters are related to the type of support and immobilization method. For example, the authors described higher kinetic parameters after invertase covalent immobilization on polymeric microbeads. Here, the result obtained may be interpreted as the difficulty for the enzyme–substrate complexes formation by the immobilized enzymes on mDE@PANI<sub>nano</sub>, most likely due to the reduced accessibility of substrates to their active sites. Microenvironment and partition effects, as well as conformational, steric and mass-transfer effects due to molecular orientation of enzyme, are well known as responsible for changes in kinetic parameters after immobilization. The  $V_{max}$  values for all immobilized derivatives were also modified and these findings could be influenced by the same factors that affected the apparent  $k_m$  constant.

A decrease in AE values for both enzymatic preparations free/immobilized: 44.42/37.86 kJ mol<sup>-1</sup> (invertase); 20.88/17.26 kJ mol<sup>-1</sup> (trypsin) was observed. This result implied increases by some orders of magnitude in their respective rate enhancements (Table 2). For invertase, although the increase in apparent  $k_m$  value after immobilization, the maintenance of optimal pH and temperature values suggest that the enzyme may have undergone conformational changes without loss of flexibility. On the other hand, a good concordance in the results was observed for immobilized trypsin since some reports [74,75] described the trypsin covalent immobilization as a process that did not caused diffusional problems and neither loss of flexibility occurred concomitantly with an increase in thermal stability. These observations could be related to interactions occurring in domains far from active site (in a region rich of lysine residues near the N-terminal hexapeptide cleaved by autolysis, for example) or to use of spacers such as glutaraldehyde offering higher mobility and avoiding steric hindrance.

$\beta$ -galactosidase is an enzyme whose rate of hydrolysis is governed by the diffusion limit of the substrate in the medium and its kinetic parameters were reported to be enhanced when displayed on nanoparticle surfaces such as quantum dots [76]. At this point, it is important to differentiate two situations: (a) the diffusion of the substrate in the surface of the immobilized derivative and; (b) the processing of the substrate into the active site of the enzyme. In the present study, immobilized  $\beta$ -galactosidase showed a shift in optimum pH value (4.5–5.0) and increases in apparent  $k_m$  and AE values (80–143 mM and



**Fig. 5.** Thermal stability of the free and immobilized enzymes tested at different temperatures (45–60 °C) and times intervals (30–120 min). The following industrial enzymes were examined: (A) invertase, (B) trypsin, and (C)  $\beta$ -galactosidase.

12.82–24.38 kJ mol<sup>-1</sup>, respectively) that are linked to a decrease in rate enhancement (Table 2) and can be ascribed to conformational changes in the active center of the enzyme and increasing stiffness.

### 3.2.3. Thermal stability

Thermal stability studies of free and immobilized enzymes (Fig. 5) were performed by enzyme incubation in the absence of substrate at different temperatures (45–60 °C) and times (30–120 min). Free invertase (Fig. 5A) exhibited thermal susceptibility at 50 °C, where it retained only approximately 25% and 10% of its activity after 30 and 60 min, respectively. In contrast, its immobilized derivative retained 86% and 76% of activity at these time points. At 55 °C, the free enzyme retained only 10% of its activity at both 30 and 60 min, whereas the immobilized form retained more than 50% of its activity. These results indicated that the thermostability of invertase increased because of covalent immobilization on mDE@PANI<sub>nano</sub>. Similar observations have been previously reported for other immobilized enzymes [64,77,78].

Trypsin exhibited the same profile in both free and immobilized forms at 45 °C (Fig. 5B). However, at 50 °C the free form lost half of its activity relative to the immobilized derivative at all time points. A

similar resistance to thermal inactivation of the immobilized derivative was also exhibited at 55 °C at 30 and 60 min, with retention of 2-fold higher activity by the immobilized trypsin relative to its free counterpart. This is consistent results of previous studies suggesting that the conformation of trypsin is stabilized by immobilization [41,79–81].

The thermal stability of  $\beta$ -galactosidase preparations (Fig. 5C) differed from that of other industrial enzymes investigated (invertase and trypsin). The profiles of free and immobilized  $\beta$ -galactosidase were similar at all temperatures and times tested. This finding was most clearly evident at 55 °C where residual activity was unaffected at all time points. When the temperature was increased to 60 °C, the immobilized derivative and the free counterpart exhibited 28% and 37% losses in activity at 30 min, respectively.

Some conditions might help to maintain the enzyme's catalytic features of the immobilized derivatives proposed in this study. The glutaraldehyde provides more than one binding sites to the enzyme structure which, in turn, acquire higher degree of conformational stiffness becoming more resistant to variations of temperature and pH and subsequently to denaturation. Furthermore, the PANI coating also contributes with a protective hydrophilic microenvironment against

denaturation due to its sponge-like structure [82]. Additionally, diatomaceous earth acts as a thermal insulator avoiding the heat from the surrounding medium which can cause an increase in temperature within the enzyme microenvironment and also preventing denaturation.

### 3.3. Reusability of the mDE@PANI-enzyme

All immobilized enzymes were used 10 times consecutively, and the percentage residual activities at each use are presented in Fig. S2 (see Supplementary material). After the 10<sup>th</sup> use, immobilized invertase retained 60% of its initial activity. Raj et al. [83] have reported an appreciable reusability of invertase immobilized on a nanogel matrix for up to seven cycles but observed a significant loss of activity (89%) in the ninth cycle. Immobilized trypsin was also reused 10 times and retained approximately 100% of the initial activity through the third cycle. Goradia et al. [84] have reported a 40% retention of immobilized trypsin activity at the sixth reuse, whereas mDE@PANI-TRYP retained 75% of its initial activity in the same cycle. Finally, immobilized  $\beta$ -galactosidase retained 100% of initial activity through four cycles and 62% after 10 cycles. Zhang et al. [85] previously used magnetic beads as a matrix to immobilize  $\beta$ -galactosidase. These authors reported that their immobilized derivative retained only 53% of the initial activity after nine cycles. All immobilized derivatives were efficient until the 10<sup>th</sup> cycle and displayed approximately 60% retention of activity; the observed decreases in activity could be attributable mainly to the loss of magnetic nanoparticles during sample processing between cycles. Washing processes with the appropriate enzyme buffer were performed fifteen times after each immobilized enzymatic use. Therefore, 150 washings were performed at the 10<sup>th</sup> reuse.

The attractive catalytic performance as well as the high reusability of all immobilized derivatives may be correlated with the superior activity retention due to the intermediate oxidation state of PANI presents in the matrix (mDE@PANI<sub>nano</sub>) as observed by Fernandes et al. [29]. In addition, no marked change in surface area of mDE nanoparticles after coating process was observed (See supplementary material, Fig. S1C-table). This feature allowed a better immobilization because of the sponge-like structure of PANI with a very large surface/volume ratio that provides a protective microenvironment against denaturation [82].

The results described here could prove to be valuable for the immobilization of different industrial enzymes using the accompanying immobilization protocol. Such results can provide important economic benefits to various industries as the three enzymes immobilized in this work represent diverse industrial applications.

## 4. Conclusion

The results obtained in this study are promising as they allow us to propose the following: (1) a simple and low-cost methodology for the synthesis of a proven magnetic composite from DE with interesting properties as a matrix for industrial enzyme immobilization; and (2) an efficient immobilization protocol by covalent binding. In addition, biochemical characterization also showed that all immobilized derivatives evidenced a good catalytic behavior and high thermal stability and reusability, making them attractive for several biotechnological applications.

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